

mammals and a small number of cave-dwelling birds.

ECLIPSE PLUMAGE In some birds, particularly waterfowl, an unobtrusive plumage adopted by the males once the breeding season is over.

ECOSYSTEM A collection of species living in the same habitat, together with their physical surroundings.

ECTOPARASITE An animal that lives parasitically on the surface of another animal's body, often by sucking its blood. Some ectoparasites spend all their lives on their hosts, but many—including fleas and ticks—develop elsewhere, and climb onto the host in order to feed.

ECTOTHERMIC Having a body temperature that is dictated principally by the temperature of the surroundings. Also known as cold-blooded.

ELYTRON (pl. ELYTRA) A hardened forewing in beetles, earwigs, and some bugs. The 2 elytra usually fit together like a case, protecting the more delicate hind wings underneath.

EMBRYO A young animal or plant in a rudimentary stage of development.

ENDOPARASITE An animal that lives parasitically inside another animal's body, either feeding directly on its tissues or stealing some of its food. Endoparasites frequently have complex life cycles, involving more than one host.

ENDOSKELETON An internal skeleton, typically made of bone. Unlike an exoskeleton, this kind of skeleton can grow in step with the rest of the body.

ENDOTHERMIC Able to maintain a constant, warm body temperature, regardless of external conditions. Also known as warm-blooded.

EPIPHYTE A plant that grows off the ground, often in the branches of trees. In the tropics, epiphytes create important microhabitats for many animals, particularly insects and amphibians.

ESTIVATION A period of dormancy in hot or dry weather. Estivation is common in animals that live in the subtropics, where there is a long dry season each year.

EVERSIBLE Capable of being turned inside out.

EXOSKELETON An external skeleton that supports and protects an animal's body. The most complex exoskeletons, formed by arthropods, consist of rigid plates that meet at flexible joints. This kind of skeleton cannot grow, and has to be shed and replaced (ecdysis) at periodic intervals. See also *Endoskeleton*.

EXTERNAL FERTILIZATION In reproduction, a form of fertilization that takes place outside the female's body, usually in water.

F

FAMILY A level used in classification. In the sequence of classification levels, a family forms part of an order, and is subdivided into one or more genera.

FEMUR (pl. FEMORA) The thigh bone in 4-limbed vertebrates. In insects, the femur is the third segment of the leg, immediately above the tibia.

FERAL Relating to an animal that comes

from domesticated stock, but that has subsequently taken up life in the wild. Common examples include city pigeons, cats, and horses.

FERTILIZATION The union of an egg cell and sperm, which creates a cell capable of developing into a new animal. In external fertilization, the process occurs outside the body (usually in water), but in internal fertilization, it takes place in the female's reproductive system.

FETUS A developing animal that is partly formed, and approaching the time when it will be born.

FILTER FEEDER An animal that eats by sieving small food items from water. Many invertebrate filter feeders, such as bivalve mollusks and sea squirts, are sessile animals that collect food by pumping water through or across their bodies. Vertebrate filter feeders, such as baleen whales, collect their food by trapping it while they are on the move.

FLAGELLUM (pl. FLAGELLA) A long, hairlike projection from a cell. A flagellum can flick from side to side, moving the cell along. Sperm cells use flagella to swim.

FLEHMEN RESPONSE A characteristic curling of the upper lip shown by many male mammals to enhance a scent by pressing air into the Jacobson's organ.

FLIGHT FEATHERS A bird's wing and tail feathers, used in flight.

FLIPPER In aquatic mammals, a paddle-shaped limb. See also *Fluke*.

FLUKE A rubbery tail flipper in whales and their relatives. Unlike the tail fins of fishes, flukes are horizontal, and they beat up and down instead of side to side.

FOOD CHAIN A food pathway that links 2 or more different species in that each forms food for the next species higher up in the chain. In land-based food chains, the first link is usually a plant. In aquatic food chains, it is usually an alga or other form of single-celled life. See also *Food web*.

FOOD WEB A collection of food chains from a particular habitat. In a food web, each species is often involved in several different food chains.

FORM A depression in the ground used by hares for concealing themselves or their young.

FOSSA (pl. FOSSAE) A pit or depression—for example, in an animal's skull.

FOSSORIAL Burrowing underground.

FRUGIVOROUS Fruit-eating.

FURCULA The forked, springlike organ attached to a springtail's abdomen.

G

GALL A tumorlike growth in a plant that is induced by an animal (typically an insect) or some other organism. By triggering the formation of galls, animals provide themselves with a safe hiding place, and a convenient source of food.

GENUS (pl. GENERA) A level used in classification. In the sequence of classification levels, a genus forms part of a family, and is subdivided into one or more species.

GESTATION PERIOD In mammals, the

interval between fertilization and birth, when the developing young is nourished by the mother via the placenta.

GILL An organ used for extracting oxygen from water. Gills are usually positioned on or near the head, or—in aquatic insects—at the end of the abdomen.

GONAD A male or female sex organ.

GRAZING Feeding on grass. See also *Browsing*.

GUARD HAIR A long hair in a mammal's coat, which projects beyond the underfur to protect it and help keep it dry.

H

HALTERES In true flies, the 2 knoblike organs that act like gyroscopes during flight, helping flies maneuver with great precision. Halteres are modified hindwings.

HEAT PITS Heat-sensitive depressions that some snakes have above the front of the upper jaw. These pits detect heat from warm-blooded animals, allowing the snakes to hunt in darkness.

HERBIVORE An animal that feeds on plants or plantlike plankton.

HERMAPHRODITE An animal that has both male and female sex organs, so it can fertilize itself. See also *Sequential hermaphrodite*.

HIBERNATION A period of dormancy in winter. During hibernation, an animal's body processes drop to a low level.

HOME RANGE The area that an animal, or group of animals, uses in its daily activities—for example, when foraging for food. See also *Territory*.

HORN In mammals, a pointed growth on the head. True horns are hollow sheaths of keratin, covering a bony horn core.

HOST An animal on or in which a parasite feeds.

HYPERPARASITE A parasite that attacks another parasite.

I

INCISOR TOOTH In mammals, a tooth at the front of the jaw, used in biting, slicing, or gnawing.

INCUBATION In birds, the period when a parent sits on the eggs and warms them, allowing them to develop. Incubation periods range from under 14 days to several months.

INSECTIVORE An animal that eats insects.

INTERNAL FERTILIZATION In reproduction, a form of fertilization that takes place inside the female's body. Internal fertilization is a characteristic of many land animals, including insects and vertebrates. See also *External fertilization*.

INTRODUCED SPECIES A species that humans have accidentally or deliberately brought into an ecosystem in which it does not naturally occur.

INVERTEBRATE An animal without a backbone.

IRRUPTION A sporadic mass movement of animals outside their normal range. Irruptions are usually short-lived, and occur in response to severe conditions, such as winter cold.

JK

JACOBSON'S ORGAN An organ in the roof of the mouth that is sensitive to airborne scents. Snakes often employ this organ to detect their prey, while some male mammals use it to find females that are ready to mate.

KEEL In birds, an enlargement of the breastbone that anchors the muscles used in flight.

KERATIN A tough structural protein found in hair, claws, and horns.

KINGDOM In classification, one of the 6 fundamental divisions of the natural world.

L

LARVA (pl. LARVAE) An immature but independent animal that looks completely different from an adult. A larva develops the adult shape by metamorphosis; in many insects, the change takes place in a resting stage that is called a pupa.

LATERAL-LINE SYSTEM A sensory system in fishes and some amphibians that detects pressure waves traveling through water, allowing the animal to sense movement. The system consists of sensors arranged in a line along both sides of the body.

LEK A communal display area used by male animals—particularly birds—during courtship. The same location is often revisited for many years.

LOPHOPHORE A crest of short tentacles found in bryozoans and some other marine invertebrates.

M

MANDIBLE The paired jaws of an arthropod, or a bone that makes up all or part of the lower jaw in vertebrates.

MANTLE In mollusks, an outer fold of skin covering the mantle cavity.

MANTLE CAVITY In mollusks, a cavity between the mantle and the rest of the body. The cavity is filled with water in aquatic mollusks, and often contains the gills.

MEDUSA In cnidarians, a bell-shaped body form, which is often free-living. During the complete life cycle, this form may alternate with a form called a polyp, which is often fixed in one place.

MELON A bulbous swelling in the heads of many toothed whales. The melon is filled with a fatty fluid, and is believed to focus the sounds used in echolocation.

METABOLISM The complete array of chemical processes that take place inside an animal's body. Some of these processes release energy by breaking down food, while others use energy by making muscles contract.

METACARPAL In 4-limbed animals, one of a set of bones in the front leg or arm, forming a joint with a digit distally. In most primates, the metacarpals form the palm of the hand.

METAMORPHOSIS A change in body shape shown by many animals—